

Internship Report

Of

**AI Technology, Digital Asset Management and Retrieval-Augmented
Generation (RAG) at Nine Triangles Digital Pvt. Ltd.**

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410031423

Name of Student: VARUN

Batch: 2024–2028

Academic Year: 2026–27

CANDIDATE'S DECLARATION

I, VARUN, Roll Number 2410031423, do hereby declare that the internship report titled “AI Technology, Digital Asset Management and Retrieval-Augmented Generation (RAG) at Nine Triangles Digital Pvt. Ltd.” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: VARUN

Roll No.: 2410031423

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout my internship at Nine Triangles Digital Pvt. Ltd. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during the course of this work. I express my sincere thanks to the AI Technology Team and the Human Resources team of Nine Triangles Digital Pvt. Ltd., Noida, for granting me the opportunity to undertake this internship from 26th May 2026 to 15th July 2026 and for their continuous mentorship on Digital Asset Management, Retrieval-Augmented Generation, prompt engineering, and AI workflow evaluation activities.

I am equally grateful to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their academic guidance, encouragement, and for providing the curricular framework that made this internship possible. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents and family for their constant blessings and support.

Signature: _____

Student Name: VARUN

Roll No.: 2410031423

INTERNSHIP COMPLETION CERTIFICATE



This is to certify that **Mr. Varun** has successfully completed his internship at **Nine Triangles Digital Pvt. Ltd., Noida**, under the **AI Technology Team**, from **26th May 2026 to 15th July 2026**.

During the internship period, he made valuable contributions to the organization by supporting AI technology initiatives, with a focus on Digital Asset Management (DAM) and Retrieval-Augmented Generation (RAG) solutions. He assisted in AI research, data organization, prompt engineering, testing AI workflows, knowledge base management, and evaluating AI-powered features to enhance the performance and usability of our technology platforms. Throughout his internship, he consistently demonstrated professionalism, dedication, analytical thinking, and a strong willingness to learn emerging AI technologies.

We appreciate his commitment and performance during his tenure with us and wish him continued success in all his future professional endeavors.

For **Nine Triangles Digital Pvt. Ltd.**

Team HR
Nine Triangles Digital Pvt. Ltd.

For NINE TRIANGLES DIGITAL PVT. LTD.

Authorised Signatory

Nine Triangles Digital Pvt Ltd
A2-912, Corporate Park, Sector 142, Noida, Uttar Pradesh 201305

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4. PROJECT DESCRIPTION

4.1 Introduction

The Bachelor of Technology programme in Computer Science and Engineering at IILM University, Greater Noida, requires every student to undertake an industry internship so that classroom learning in computing, data structures, and emerging technologies can be tested against the practical demands of a professional environment. This report documents the internship undertaken by VARUN (Roll No. 2410031423, B.Tech CSE, Batch 2024–2028) at Nine Triangles Digital Pvt. Ltd., Noida, from 26th May 2026 to 15th July 2026, under the organization's AI Technology Team, in the role of Management Trainee.

Artificial Intelligence has moved rapidly from a research discipline into an operational capability that organizations use to manage information, accelerate decision-making, and improve the usability of their digital platforms. Two ideas sit at the centre of this shift. The first is Digital Asset Management (DAM), the practice of organizing, tagging, storing, and retrieving an organization's digital content – documents, creative assets, and structured records – so that it can be found and reused efficiently. The second is Retrieval-Augmented Generation (RAG), an architecture that pairs a Large Language Model (LLM) with a retrieval mechanism over an organization's own knowledge base, so that AI-generated responses are grounded in accurate, current, and organization-specific information rather than in the model's general training data alone. Together, DAM and RAG describe a pipeline in which well-organized digital assets become the raw material for reliable, context-aware AI systems.

As per the internship completion certificate issued by Nine Triangles Digital Pvt. Ltd. (reproduced in Section 3 and Section 4.10 of this report), my internship involved supporting the organization's AI technology initiatives with a specific focus on Digital Asset Management and Retrieval-Augmented Generation solutions. My documented activities included assisting in AI research, data organization, prompt engineering, testing of AI workflows, knowledge-base management, and evaluation of AI-powered features intended to enhance the performance and usability of the organization's technology platforms.

This report is organized into the Project Description chapter, which explains the organization, the problem area, the objectives and scope of the internship, the technologies and concepts involved, the system architecture and methodology that frame this domain, the outcomes of the internship, and the supporting certificates and communication records, followed by a Bibliography chapter that lists the academic and industry sources referred to while preparing this report. Throughout, general technical explanations of AI, DAM, and RAG concepts are presented separately from specific

claims about the activities I personally carried out, so that the report remains an accurate record of the internship.

4.2 Organization Profile

Nine Triangles Digital Pvt. Ltd. is a boutique digital strategy consulting and integrated-implementation company headquartered at A2-912, Corporate Park, Sector 142, Noida, Uttar Pradesh – 201305, India. The company was founded in 2017 by Krishna Mohan Jha and positions itself as a strategy-first digital consultancy that takes ownership of end-to-end delivery across four practice areas: Creative, Technology, Media, and Data Analytics. According to information published on the company's own website and professional business directories, Nine Triangles has delivered digital transformation, campaign, and technology engagements for clients across banking, financial services, hospitality, publishing, and education sectors, including organizations such as ICICI Bank, ICICI Lombard, DSP Mutual Fund, and Newgen Software.

Within its Technology practice, the organization has stated a capability to design data-driven and machine-learning-aided solutions so that client brands can gain an innovative edge across competitive and industry contexts. My internship was situated within this Technology practice, under the AI Technology Team, which focuses on applying AI methods – including Digital Asset Management and Retrieval-Augmented Generation – to improve how the organization and its stakeholders manage and use information.

All organizational details in this profile are drawn from publicly available company information and from my own offer letter and internship completion certificate; no proprietary financial, client-specific, or internal operational information has been included, in keeping with the confidentiality expectations set out in my offer letter.

4.3 Problem Statement

Modern organizations generate and accumulate large volumes of digital assets – documents, creative files, research notes, process records, and communication artefacts – far faster than traditional, manually maintained folder structures can organize them. When these assets are poorly tagged, duplicated, or scattered across disconnected systems, employees spend disproportionate time searching for information instead of using it, and AI systems built on top of this information inherit the same disorganization, producing answers that are incomplete, outdated, or ungrounded in the organization's actual knowledge.

The problem area addressed during this internship was, therefore, twofold: first, how digital assets and organizational knowledge can be structured and maintained so that

they form a reliable knowledge base; and second, how that knowledge base can be connected to AI/LLM-based tools – through retrieval, prompt engineering, and systematic evaluation – so that AI-powered features built on top of it remain accurate, relevant, and genuinely useful to the people who rely on the organization's technology platforms. This report focuses on the concepts, workflow, and evaluation practices relevant to this problem area, to the extent that they are supported by my documented internship activities.

4.4 Project Objectives

- To understand, at a working level, how Digital Asset Management practices organize and govern an organization's digital content.
- To study the principles of Retrieval-Augmented Generation and how retrieval and generation stages combine to produce grounded AI responses.
- To assist in organizing data and digital assets so that they are more easily discoverable and reusable.
- To gain hands-on exposure to prompt engineering as a technique for improving the quality and reliability of AI-generated output.
- To assist in testing AI workflows and identifying gaps between expected and actual system behaviour.
- To support knowledge-base management activities that keep organizational information current and well-structured.
- To assist in evaluating AI-powered features against usability and performance considerations, and to document findings in a structured manner.
- To develop a professional understanding of how an AI Technology team operates within a digital consulting organization.

Table 4.2: Mapping of Certificate-Supported Activities to Objectives

Certificate-Supported Activity	Related Objective(s) / Report Section
AI research (certificate-supported)	Objective 1, 2 – Sections 4.4, 4.7
Data organization	Objective 3 – Sections 4.4, 4.6
Prompt engineering	Objective 4 – Sections 4.4, 4.8
Testing AI workflows	Objective 5 – Sections 4.4, 4.8
Knowledge-base management	Objective 6 – Sections 4.4, 4.7
Evaluating AI-powered features	Objective 7 – Sections 4.4, 4.8, 4.9

4.5 Scope of the Project

The scope of this internship was applied and evaluative in nature rather than product-development in nature. It covered exposure to, and assistance with, AI research, data and knowledge-base organization, prompt engineering, AI workflow testing, and feature evaluation, carried out under the supervision of the AI Technology Team at Nine Triangles Digital Pvt. Ltd. between 26th May 2026 and 15th July 2026. The internship did not extend to the design or deployment of a proprietary production system, and this report does not claim ownership of any specific commercial architecture, dataset, or software product belonging to the organization or its clients. Where the report discusses AI, DAM, and RAG concepts in technical depth, this is intended as an academic explanation of the domain in which the internship took place, and is clearly distinguished from the specific, certificate-supported activities carried out personally.

4.6 Technologies and Tools Used

The internship activities were framed around the following technology concepts. In line with the scope defined above, these are presented as general/conceptual technologies relevant to the domain rather than as a list of specific proprietary software products, since no such product names are attested in my internship documentation.

Table 4.1: Technologies and Tools / Concepts Used

Concept / Technology	Description
Artificial Intelligence (AI)	Broad field concerned with building systems that perform tasks normally requiring human intelligence, such as understanding language and retrieving relevant information.
Large Language Models (LLMs)	AI models trained on large text corpora that generate natural-language responses; the generation stage of a RAG pipeline.
Retrieval-Augmented Generation (RAG)	An architecture that augments an LLM's prompt with content retrieved from an external knowledge base, improving factual grounding.
Digital Asset Management (DAM)	Practices and systems for organizing, tagging, storing, and retrieving an organization's digital content.
Prompt Engineering	The practice of designing and refining input prompts to improve the accuracy and relevance of LLM output.
Knowledge Base Management	Maintaining a structured, current repository of organizational information that AI systems and employees can draw upon.
Data Organization / Tagging	Structuring raw digital assets with consistent metadata, categories, and naming conventions to support retrieval.

Concept / Technology	Description
AI Workflow Testing & Evaluation	Systematic checking of AI-powered features against expected behaviour, relevance, and usability criteria.

4.7 System Architecture

Although this internship did not involve building a proprietary production system, understanding the generic architecture that underlies AI-powered DAM/RAG solutions is central to the work carried out. Figure 4.1 illustrates a representative, technology-agnostic workflow of this kind, rather than any specific architecture owned by Nine Triangles Digital Pvt. Ltd. or its clients.

The workflow begins with an organization's digital assets – documents, media, and structured records. These assets are organized into a knowledge base, which involves classifying, tagging, and structuring content so that it can be indexed meaningfully. The data-processing and indexing stage converts this organized content into a form (such as embeddings or structured indices) that supports efficient similarity search. When a user submits a query, the retrieval layer searches this index and returns the most relevant context, which is then combined with the user's query to construct an augmented prompt. This prompt is passed to a Large Language Model, which generates a response grounded in the retrieved context rather than relying solely on its pre-trained knowledge. Finally, the generated response passes through an evaluation and validation stage, where its relevance, accuracy, and usefulness are assessed; insights from this stage feed back into the indexing and knowledge-base layers, forming a continuous improvement loop, which is the essence of AI workflow testing and feature evaluation.

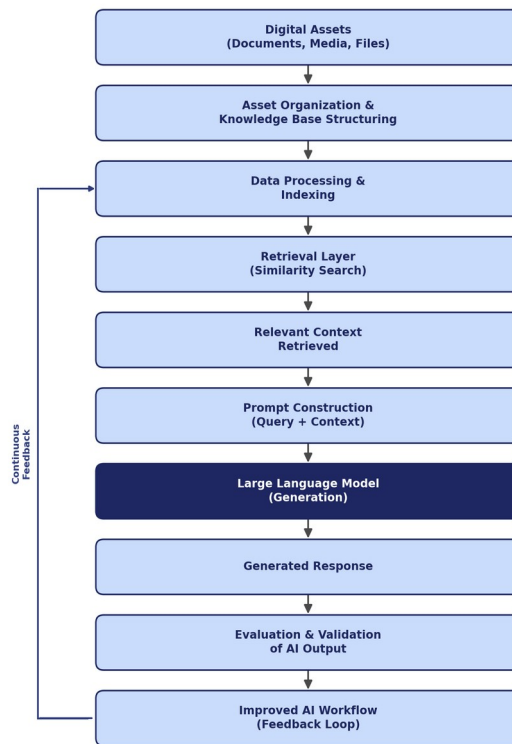


Fig. 4.1: Generic AI / DAM-RAG Workflow Architecture

Within this architecture, the key challenges that AI Technology teams typically address include keeping the knowledge base current as assets change, avoiding retrieval of irrelevant or stale context, controlling for hallucination in the generation stage, and defining evaluation criteria that meaningfully capture whether a response is both correct and usable. These challenges directly motivated the emphasis, in my internship, on knowledge-base management, prompt engineering, and structured AI workflow testing.

4.8 Methodology

The internship followed a structured, progressively deepening approach consistent with the activities documented in my internship completion certificate. Figure 4.2 summarizes this approach as a set of broad stages over the seven-week internship period.

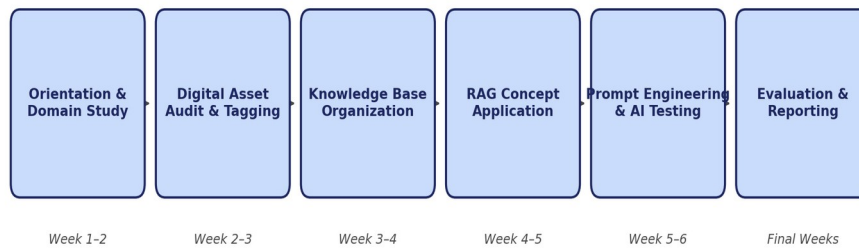


Fig. 4.2: Weekly Internship Methodology

During the orientation stage, I was introduced to the AI Technology Team's working domain, including the fundamentals of Digital Asset Management and the role of Retrieval-Augmented Generation in improving AI system reliability. This was followed by hands-on exposure to digital asset audit and tagging practices, where existing content was reviewed for completeness and consistency of metadata.

The knowledge-base organization stage involved structuring information so that it could support efficient retrieval – grouping related content, standardizing terminology, and removing duplication. Building on this foundation, the RAG-concept-application stage focused on understanding how retrieval and generation interact in practice, including how the quality of retrieved context influences the quality of generated answers.

The prompt-engineering and AI-testing stage was the most iterative part of the methodology: prompts were drafted, tested against representative queries, and refined based on the relevance and accuracy of the resulting output, following an iterate-observe-refine cycle common to prompt-engineering practice. Throughout this stage, AI workflows were exercised with varied inputs to surface edge cases and inconsistencies. In the final weeks, the emphasis shifted to evaluation – assessing AI-powered features against usability and performance considerations – and to consolidating findings and documentation, culminating in this report.

4.9 Expected Outcomes

The internship resulted in the following learning and professional outcomes, consistent with the activities recorded in my internship completion certificate:

- A working understanding of Digital Asset Management practices and their role in supporting downstream AI systems.
- A practical, applied understanding of Retrieval-Augmented Generation, including the relationship between retrieval quality and generation quality.

- Improved data-organization skills, including tagging, structuring, and de-duplicating digital content.
- Hands-on prompt-engineering experience, including iterative refinement of prompts against observed output quality.
- Exposure to structured AI workflow testing methods and the kinds of failure modes such testing is designed to catch.
- Familiarity with knowledge-base management practices that keep organizational information usable over time.
- Experience evaluating AI-powered features against usability and performance criteria, and documenting the results professionally.
- Broader professional outcomes, including analytical thinking, collaborative working within a technology team, and clear technical communication – qualities specifically noted in my internship completion certificate.

These outcomes are presented as the natural results of the certificate-supported activities described above; no specific quantitative performance metrics, proprietary results, or client outcomes are claimed, as none are attested in my supporting documentation.

4.10 Certificates of Completion and Communication Proof

This section reproduces the official communication and certification documents issued to me by Nine Triangles Digital Pvt. Ltd. in relation to this internship: the offer letter confirming my engagement as Management Trainee, and the internship completion certificate confirming successful completion of the internship under the AI Technology Team.



22nd May'2026
Noida

Sub: Offer Letter for Management Trainee

Dear Mr. Varun

Based on the details provided by you during the interview process, we are happy to offer you the position of Management Trainee – Technology Services with our organisation. Your management trainee offer details are as follows:

Position	Management Trainee – Creative Services
Cost-to-Company	Nil

You are requested to report on Tuesday, 26th May 2026

Terms and Conditions:

1. You would be expected to abide by all the rules, regulations, values and systems existing in the Company along with offering complete respect to the trade secrets and brand of Nine Triangles Digital.
2. The employee will be completely and solely responsible for portable infrastructure (laptop computers, mobile phones, company owned vehicles, etc.) being used by him/her during the course of his/her work with the company. The company would be entitled to recover any damage caused to the above from the employee.

You are requested to send confirmation of acceptance of the above offer by today's end of the day. Please feel free to revert for any further information / assistance to facilitate your joining process.

We wish you a satisfying, enriching and lasting association with Nine Triangles Digital.

Regards,

Team HR
Nine Triangles Digital Pvt. Ltd.

For NINE TRIANGLES DIGITAL PVT. LTD.

Authorised Signatory

Nine Triangles Digital Pvt. Ltd.
A2-912, Corporate Park, Sector 142, Noida - 201305
INDIA

Offer Letter – Nine Triangles Digital Pvt. Ltd. (dated 22nd May 2026)



This is to certify that **Mr. Varun** has successfully completed his internship at **Nine Triangles Digital Pvt. Ltd., Noida**, under the **AI Technology Team**, from **26th May 2026 to 15th July 2026**.

During the internship period, he made valuable contributions to the organization by supporting AI technology initiatives, with a focus on Digital Asset Management (DAM) and Retrieval-Augmented Generation (RAG) solutions. He assisted in AI research, data organization, prompt engineering, testing AI workflows, knowledge base management, and evaluating AI-powered features to enhance the performance and usability of our technology platforms. Throughout his internship, he consistently demonstrated professionalism, dedication, analytical thinking, and a strong willingness to learn emerging AI technologies.

We appreciate his commitment and performance during his tenure with us and wish him continued success in all his future professional endeavors.

For **Nine Triangles Digital Pvt. Ltd.**

Team HR
Nine Triangles Digital Pvt. Ltd.

For NINE TRIANGLES DIGITAL PVT. LTD.

Authorized Signatory

Nine Triangles Digital Pvt Ltd
A2-912, Corporate Park, Sector 142, Noida, Uttar Pradesh 201305

Internship Completion Certificate – Nine Triangles Digital Pvt. Ltd.

5. BIBLIOGRAPHY / REFERENCES

References are listed alphabetically by author/organization, in single line spacing as per the prescribed report format.

- [1] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems (NeurIPS)*, 33.
- [2] AIIM – The Association for Intelligent Information Management. Digital Asset Management: Overview and Best Practices. Retrieved from <https://www.aiim.org>
- [3] IBM. What is Retrieval-Augmented Generation (RAG)? IBM Think Topics. Retrieved from <https://www.ibm.com/topics/retrieval-augmented-generation>
- [4] Google Cloud. Prompt Engineering Overview. Retrieved from <https://cloud.google.com/discover/what-is-prompt-engineering>
- [5] Nine Triangles Digital Pvt. Ltd. Company Profile. Retrieved from <https://www.ninetriangles.com>
- [6] Nine Triangles Digital Pvt. Ltd. (2026). Offer Letter issued to Varun, dated 22nd May 2026.
- [7] Nine Triangles Digital Pvt. Ltd. (2026). Internship Completion Certificate issued to Mr. Varun, AI Technology Team.
- [8] IILM University, Greater Noida. Internship Report Format Guidelines, Academic Year 2026–27 (School of Computer Science and Engineering).
- [9] OpenAI. Best Practices for Prompt Engineering. Retrieved from <https://platform.openai.com/docs>
- [10] Microsoft Learn. What is Retrieval Augmented Generation (RAG)? Retrieved from <https://learn.microsoft.com>